observation is an integral element of quantum mechanics, being the catalyst that coaxes one outcome from the quantum haze to be realized.[9] Personally, I find this extremely unlikely, but I know of no way to rule it out.

The upshot is that the status of free will and its role within fundamental physical law remain unresolved. So let's consider both possibilities, free will that's illusory and free will that's real.

If free will is an illusion, and if time travel to the past is possible, then your inability to prevent your parents from meeting poses no puzzle. Although you feel as if you have control over your actions, the laws of physics are really pulling the strings. When you go to whisk away your mother or shoot your father, the laws of physics get in the way. The time machine lands you on the wrong side of town, and you arrive after your parents have met; or you try to pull the trigger and the gun jams; or you do pull the trigger, but you miss the target and instead knock off your father's only competitor for your mother's hand, clearing the way for their union; or, perhaps, when you step out of the time machine you no longer have the desire to prevent your parents from meeting. Regardless of your intention when you enter the time machine, your actions when you exit are part of spacetime's consistent story. The laws of physics trump all attempts to thwart logic. Everything you do fits in perfectly. It always has and always will. You can't change the unchangeable.

If free will is not an illusion, and if time travel to the past is possible, quantum physics gives alternative suggestions for what might happen, and is distinctly different from the formulation based on classical physics. One particularly compelling proposal, championed by Deutsch, makes use of the Many Worlds interpretation of quantum mechanics. Remember from Chapter 7 that in the Many Worlds framework, every potential outcome embodied in a quantum wavefunction—a particle's spinning this way or that, another particle's being here or there—is realized in its own separate, parallel universe. The universe we're aware of at any given moment is but one of an infinite number in which every possible evolution allowed by quantum physics is separately realized. In this framework, it's tempting to suggest that the freedom we feel to make this or that choice reflects the possibility we have to enter this or that parallel universe in a subsequent moment. Of course, since infinitely